

Homework: Lagrangians, Hessians, and Bordered Hessians

GSE 510

Due: Next Class

Instructions: Answer all questions carefully. Show your work. For constrained optimization problems, clearly write the Lagrangian, compute the first-order conditions, solve for the candidate optimum, and use the appropriate second-order condition when asked. When the problem has nonnegativity constraints, remember to check whether the Lagrangian solution is actually feasible and economically meaningful.

Practice Problems: Work together in class

1. Consider the unconstrained function

$$f(x, y) = 20 + 6x + 8y - x^2 - 2y^2.$$

- (a) Compute f_x and f_y .
- (b) Find the critical point.
- (c) Compute the Hessian matrix.
- (d) Use the Hessian to determine whether the critical point is a local maximum, local minimum, or saddle point.
- (e) Find the value of the function at the critical point.

2. Consider the constrained maximization problem

$$\max_{x, y} xy$$

subject to

$$x + y = 10.$$

- (a) Write the Lagrangian.
- (b) Compute the first-order conditions.
- (c) Solve for x^* , y^* , and λ^* .
- (d) Interpret the solution geometrically.
- (e) What is the maximum value of xy ?

3. Consider the consumer problem

$$\max_{x, y} x^{1/2} y^{1/2}$$

subject to

$$2x + y = 12.$$

- (a) Write the Lagrangian.
- (b) Compute the first-order conditions.
- (c) Use the first-order conditions to derive the tangency condition.
- (d) Solve for the optimal bundle (x^*, y^*) .

- (e) Find the value of the Lagrange multiplier λ^* .

4. Consider the problem

$$\max_{x,y} f(x,y) = xy$$

subject to

$$x + y = 10.$$

The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = xy + \lambda(10 - x - y).$$

- (a) Compute the Hessian matrix of the objective function $f(x, y)$.
- (b) Compute the bordered Hessian for the constrained problem.
- (c) Evaluate the bordered Hessian at the critical point.
- (d) Use the bordered Hessian condition to determine whether the critical point is a constrained maximum or minimum.
- (e) Compare your answer with your intuition from Question 2.

5. Consider the constrained minimization problem

$$\min_{x,y} x^2 + y^2$$

subject to

$$x + y = 6.$$

- (a) Write the Lagrangian.
- (b) Compute the first-order conditions.
- (c) Solve for x^* and y^* .
- (d) Compute the bordered Hessian.
- (e) Use the bordered Hessian to determine whether the solution is a constrained minimum.

Homework Problems: Submit individually

6. Consider the unconstrained function

$$f(x, y) = 4x^2 + 2y^2 - 4xy - 8x - 4y.$$

- (a) Compute f_x and f_y .
- (b) Find the critical point.
- (c) Compute the Hessian matrix.
- (d) Compute the determinant of the Hessian.
- (e) Classify the critical point.
- (f) Find the value of the function at the critical point.

7. Consider the consumer problem

$$\max_{x, y} x^{1/3} y^{2/3}$$

subject to

$$x + 2y = 18.$$

- (a) Write the Lagrangian.
- (b) Compute the first-order conditions.
- (c) Use the first-order conditions to derive the relationship between x and y at the optimum.
- (d) Solve for the optimal bundle (x^*, y^*) .
- (e) Find the maximum utility.
- (f) Interpret the Lagrange multiplier in words.

8. Consider the firm problem

$$\max_{L, K} 40L + 60K - L^2 - 2K^2 - LK$$

subject to

$$L + K = 20.$$

- (a) Write the Lagrangian.
- (b) Compute the first-order conditions.
- (c) Solve for L^* , K^* , and λ^* .
- (d) Compute the Hessian matrix of the objective function.
- (e) Compute the bordered Hessian for the constrained problem.
- (f) Use the bordered Hessian to determine whether the critical point is a constrained maximum.
- (g) Interpret the economic meaning of the constraint.

9. **A case where the Lagrangian answer is not the actual economic solution.**

A consumer has utility

$$u(x, y) = x + 2y$$

and budget constraint

$$x + y = 10,$$

with nonnegativity constraints

$$x \geq 0, \quad y \geq 0.$$

- (a) Write the Lagrangian for the equality-constrained problem, ignoring the nonnegativity constraints.
- (b) Compute the first-order conditions.
- (c) Explain why the usual interior Lagrangian method does not give a meaningful interior solution.
- (d) Evaluate utility at the two endpoints of the budget line:

$$(10, 0) \quad \text{and} \quad (0, 10).$$
- (e) What is the actual utility-maximizing bundle?
- (f) Explain why this is a corner solution.
- (g) What lesson does this example teach about using Lagrangians in economic problems?

10. **Slightly harder problem.** Consider the constrained optimization problem

$$\max_{x,y} f(x,y) = x^2y$$

subject to

$$x + y = 12,$$

with

$$x \geq 0, \quad y \geq 0.$$

- (a) Write the Lagrangian.
- (b) Compute the first-order conditions for an interior solution.
- (c) Solve for the interior critical point.
- (d) Evaluate the objective function at the interior critical point.
- (e) Evaluate the objective function at the endpoints $(12, 0)$ and $(0, 12)$.
- (f) Is the interior critical point the global maximum on the feasible set?
- (g) Compute the bordered Hessian at the interior critical point.
- (h) Explain why checking the boundary matters in constrained optimization.

Reminder: The Hessian and Bordered Hessian

For an unconstrained problem with objective function $f(x, y)$, the Hessian is

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

At a critical point:

- If $|H| > 0$ and $f_{xx} < 0$, the point is a local maximum.
- If $|H| > 0$ and $f_{xx} > 0$, the point is a local minimum.
- If $|H| < 0$, the point is a saddle point.
- If $|H| = 0$, the test is inconclusive.

For a constrained problem with one equality constraint

$$g(x, y) = c,$$

the Lagrangian can be written as

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda(c - g(x, y)).$$

The bordered Hessian is

$$B = \begin{bmatrix} 0 & g_x & g_y \\ g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ g_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{bmatrix}.$$

For two choice variables and one equality constraint, the sign of the determinant of the bordered Hessian helps classify the constrained critical point. In the problems above, use the sign convention discussed in class.